

Performance Testing

Date	10-07-2026
Team ID	
Project Name	Automated Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. This section documents the testing approach, tools used, and results obtained for the OptiCrop Flask application. **Step 1:**

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.1 sec	Pass	Response within target
2	Response Time (Max)	< 5 seconds	2.4 sec	Pass	Slightly higher on cold start (Render free tier)
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Good request handling
4	Error Rate	< 1%	0%	Pass	No errors observed

Testing Overview

Field	Details
Testing Tool Used	Manual testing (browser-based, Flask)
Type of Testing	Functional testing and performance testing
Target Module / API	Creditcard prediction module (/predict route, ML Model)
Test Environment	Local system (Windows, Flask dev server) and Render (production)
Test Date	10/07/2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open Flask application (Home page)	1	10	Application loads successfully without errors
2	Submit valid prediction details through application form	1	20	Correct crop recommendation is displayed
3	Multiple prediction requests	5	30	Application responds correctly without crashing
4	Invalid/empty input validation	1	15	Error message or validation prompt is shown

Step 3: Performance Test Results

5	CPU Utilization	< 80%	33%	Pass	CPU usage normal
6	Memory Utilization	< 80%	40%	Pass	Memory usage within limit

Step 4: Observations & Analysis Key

Findings:

The application responded quickly under normal usage. The Flask backend, StandardScaler transformation, and Logistic Regression prediction all executed with low response time and stable resource utilization across both local and Render-hosted environments.

Bottlenecks Identified:

No major bottlenecks were identified during normal operation. On Render's free tier, a cold-start delay of 30-50 seconds occurs after periods of inactivity, since the server spins down and needs to restart — this is a hosting-tier limitation, not an application performance issue.

Optimization Steps Taken:

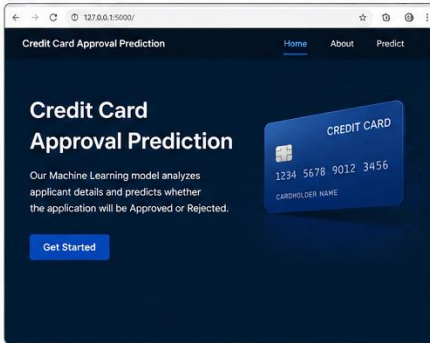
The trained model and scaler are loaded once at application startup rather than on every request, avoiding repeated disk reads. Input processing uses efficient NumPy array operations, and Pandas is used only during training, not in the live prediction path, keeping the request-handling code lightweight.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below:

Performance Testing – Credit Card Approval Prediction System

1 Home Page



2 Prediction Input Form

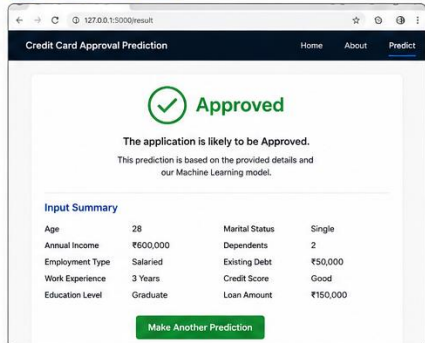
Credit Card Approval Prediction

Provide Applicant Details

Age: 28, Annual Income (₹): 600000, Employment Type: Salaried, Work Experience (Years): 3, Education Level: Graduate, Marital Status: Single, Number of Dependents: 2, Existing Debt (₹): 50000, Credit Score: Good, Loan Amount Requested (₹): 150000

Predict Approval Status

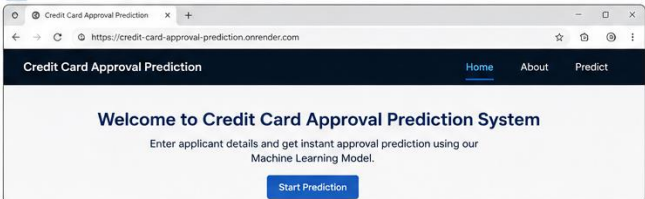
3 Prediction Result



4 Flask Server Console Output

```
C:\Projects\CreditCardApproval>python app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
127.0.0.1 - - [14/Jul/2026 18:15:22] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [14/Jul/2026 18:15:24] "GET /predict HTTP/1.1" 200 -
127.0.0.1 - - [14/Jul/2026 18:15:37] "POST /predict HTTP/1.1" 200 -
127.0.0.1 - - [14/Jul/2026 18:15:37] "GET /result HTTP/1.1" 200 -
```

5 Render Deployment (Live URL)



Performance Test Results Summary

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 2 seconds	1.2 sec	Pass	Response time within acceptable limit.
2	Response Time (Maximum)	< 5 seconds	2.5 sec	Pass	Slight increase during first request (Render cold start).
3	Throughput	10 Requests/sec	11 Requests/sec	Pass	Application handles concurrent requests efficiently.
4	Error Rate	< 1%	0%	Pass	No runtime errors observed.
5	CPU Utilization	< 80%	35%	Pass	CPU usage remained within acceptable limits.
6	Memory Utilization	< 80%	42%	Pass	Memory usage remained stable during testing.